Experiment 7: Law makers - Henry and Boyle

Notes

The 'bends' or decompression sickness is most commonly associated with deep-sea divers. When a diver comes to the surface too quickly from a dive it can cause inert gases, usually nitrogen, that are normally dissolved in body fluids and tissues, to come out of solution (outgassing) and form gas bubbles. I am sure that you have observed this happen when you have opened a can or bottle of carbonated drink like Coke® or Pepsi®; you both hear and see the bubbles of gas forming in the liquid. The "bends" is also a hazard for crew and passengers in high-flying aircraft as cabin pressures may drop to as low as 73% of sea level pressure, but more particularly if the cabin pressurisation system fails.

In 1803 William Henry explained the phenomenon responsible for decompression sickness or 'the bends'. Today his explanation is known as Henry's Law and states: "when the pressure of a gas over a liquid is decreased, the amount of gas dissolved in that liquid will also decrease."

Mathematically the relationship can be described as:

$$\mathbf{p} \ \alpha \ \mathbf{c}$$

$$\mathbf{p} = \mathbf{kc}$$

where: **p** - is the partial pressure of the solute gas above the solution;
c - is the concentration of the solute gas in the solution;
k - is the Henry's Law constant.

Henry's Law describes the relationship between pressure and gas solubility.

Irish chemist Robert Boyle in 1662, a lot earlier than Henry, investigated the relationship between gas pressure and gas volume. He too has his own law that summarised his findings.

In this experiment you will investigate Boyle's Law. The forces between gas molecules are relatively weak, so the molecules have very little effect on one another and the gas molecules are so far apart that the volume of the gas molecules is negligable compared to the distance between them. For this reason the relationship observed between the pressure and volume of a sample of air will be general for all gases. A fixed amount of air trapped in a syringe is used and the pressure applied to the gas is increased, by pushing in the syringe. The volume of air will be recorded for each value of pressure. You will graph the values of pressure and volume to determine a mathematical relationship between pressure and volume.

This experiment may be performed either by using a graduated syringe with a sealed end and a set of masses (Method A) or by using a pressure sensor (Method B) that is interfaced to a computer or a graphic calculator.

Equipment

Method A: using a graduated syringe

graduated syringe with sealed end - about 50 mL
set of masses (six 1 kg or twelve 500 g masses)
stand and clamp
balance

Procedure

Method A: using a graduated syringe

1. Make a table of your results as shown below.

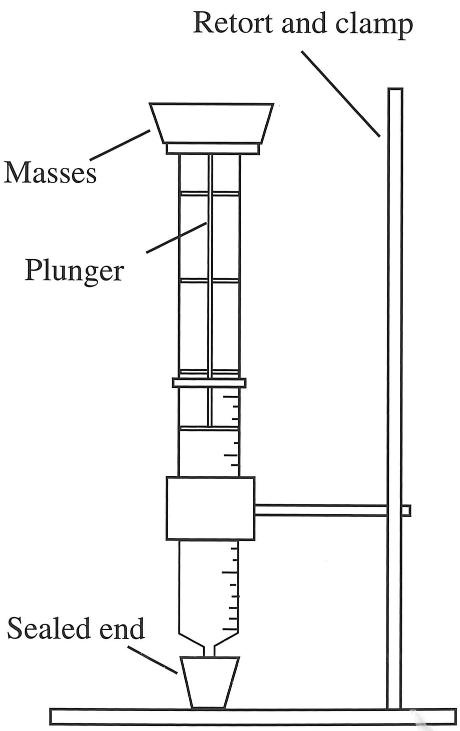
Pressure due to masses (kg)	Volume (mL)			Average volume (mL)	Total pressure (kg)	$\frac{1}{V}$ (mL^{-1})	PV
	Trial 1	Trial 2	Trial 3				
0							
1							
2							

2. Weigh the plunger on an accurate balance and record its mass.

3. Adjust the plunger so that it is just below the highest calibration on the syringe and ensure that the syringe is air-tight. Clamp the syringe in an upright position and read the volume of air trapped in the syringe as precisely as possible.

4. Carefully place a kilogram mass on top of the plunger and again read the volume of air in the syringe. Remove the mass and place it back on twice more so as to record three volume measurements.

5. Repeat procedure # 3 using a mass of 2 kg, and continue increasing the masses until a mass of 6 kg has been added.



SAFETY NOTE:
- **The syringe may fail under pressure.**
- **Wear safety glasses.**
- **Use a method to support the weights so that they cannot topple from the top of the syringe.**

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Processing of results and questions

Method A: using a graduated syringe

To calculate the pressure acting on the gas in the syringe it is necessary to take account of atmospheric pressure, the mass of the plunger and the masses placed on top of the plunger. You will use the sum of the masses of the plunger, the added masses and the air column above the syringe as a measure of the pressure exerted on the gas. This is a reasonable procedure because all these masses experience the same gravitational acceleration and act on the same cross-sectional area of the syringe.

1. To work out the mass of air acting on the syringe you need to multiply the mass per unit area of an air column above the earth (1.03 kg cm^{-2}) by the cross sectional area of the syringe.

   mass of air acting on syringe
   =(mass/unit area of air column) × *(cross sectional area of syringe)*
   = (1.03 kg cm^{-2}) × (πr^2)

2. Calculate the total pressure (in kilograms) in each case by working out the total mass acting on the syringe. Make sure all your masses are expressed in kilograms. Record these values in the table.

   pressure = mass of air column + mass of plunger + added masses (in kg)

3. Using the data from the various trials, determine the average gas volume for each pressure and record these values in the table.

4. Draw a graph of these results, plotting the pressure (in kilograms) on the vertical axis and the average volume (in millilitres) on the horizontal axis. Draw a smooth curve of best fit through the points.

5. Use the average volumes to determine values of $\frac{1}{V}$. Record these values, to the appropriate number of significant figures, in the table.

6. Draw another graph, this time plotting the pressure (in kilograms) on the vertical axis and values of $\frac{1}{V}$ on the horizontal axis. Rule a line of best fit through the points.

7. Calculate the values of the products of total pressure and average volume and record these in the table.

8. From your results what can you conclude about the relationship between the pressure and volume of a gas?

9. Why do you think that the values of the PV products were not exactly the same?

10. Under what circumstances would the gas volume approach zero?

Equipment

Method B: using a pressure sensor

computer or graphic calculator loaded with suitable software
computer or graphic calculator interface
pressure sensor
graduated syringe with sealed end

Procedure

Method B: using a pressure sensor

1. Ensure that the pressure sensor is connected to the interface and that the interface is connected to the computer or graphic calculator.
2. Check that the program has been loaded and has been set up with pressure on the vertical axis and volume on the horizontal axis.
3. Check that the pressure sensor has been calibrated. If this needs to be done, follow the software instructions. Assume atmospheric pressure is 1.00 atmosphere. Now take pressure-volume readings following the procedures in steps 4 and 5.
4. With the air valve open, adjust the syringe to its largest volume. Close the air valve and enter on the screen the volume of gas in the syringe. Also enter the air pressure reading of 1.00 atmosphere.
5. Take at least four more readings. Do this by pushing the syringe in to decrease the volume. Hold the syringe steady and record the pressure and volume readings. Repeat this until you have at least five pressure-volume readings.
6. Examine the graph of pressure versus volume. Decide whether the relationship is direct or inverse. Then, using the curve of best fit procedure, draw the graph.
7. Obtain printouts of the graph and the data table.

Processing of results and questions

Method B: using a pressure sensor

1. Use the values from the data table to calculate PV (pressure x volume) for each pair of readings and comment on the results.
2. Using a computer spreadsheet or a graphic calculator, plot pressure (P) versus the reciprocal of volume ($\frac{1}{V}$) to confirm the relationship between pressure and volume. To do this you will to create a new data column ($\frac{1}{V}$). Enter the formula ($\frac{1}{V}$) and the appropriate units (mL^{-1}). Change the axis of the graph and print the graph.
3. What conclusion about the relationship between the pressure and volume of a gas can you make from the results?
4. Under what conditions would it be possible to obtain a gas volume that approaches zero?

Notes